

Coding Patterns — Practice These, Not 1000 Random LeetCode Problems

Instead of grinding random problems, master these **21 core patterns**. Most interview problems are just variations of one of these.

Legend

- Each pattern covers a **specific problem-solving template**
- 5 LeetCode problems per pattern — from easy warm-up to hard stretch
- Difficulty: ● Easy · ● Medium · ● Hard

The Patterns

#	Pattern	When to Use	Problem 1	Problem 2	Problem 3	Problem 4	Problem 5
1	Two Pointers	Sorted array/string; find pairs, triplets, or squeeze from both ends	● Valid Palindrome #125	● Two Sum II #167	● 3Sum #15	● Container With Most Water #11	● Trapping Rain Water #42
2	Sliding Window	Contiguous subarray/substring with a constraint (size, sum, unique chars)	● Longest Substring Without Repeating Characters #3	● Permutation in String #567	● Longest Repeating Character Replacement #424	● Minimum Window Substring #76	● Sliding Window Maximum #239
3	Fast & Slow Pointers	Cycle detection in linked list or array; find middle of list	● Linked List Cycle #141	● Middle of the Linked List #876	● Linked List Cycle II #142	● Happy Number #202	● Find the Duplicate Number #287
4	Merge Intervals	Overlapping intervals; scheduling; range merging	● Merge Intervals #56	● Insert Interval #57	● Non-overlapping Intervals #435	● Meeting Rooms II #253	● Employee Free Time #759
5	Cyclic Sort	Array contains numbers in range [1, n]; find missing/duplicate	● Missing Number #268	● Find the Duplicate Number #287	● Find All Numbers Disappeared in an Array #448	● Find All Duplicates in an Array #442	● First Missing Positive #41

6	In-place Reversal of Linked List	Reverse all or part of a linked list without extra space	 Reverse Linked List #206	 Reverse Linked List II #92	 Palindrome Linked List #234	 Reorder List #143	 Reverse Nodes in k-Group #25
7	Tree BFS (Level Order)	Level-by-level processing; shortest path in unweighted tree	 Symmetric Tree #101	 Binary Tree Level Order Traversal #102	 Binary Tree Right Side View #199	 Binary Tree Zigzag Level Order Traversal #103	 Populating Next Right Pointers in Each Node #116
8	Tree DFS (Path Problems)	Root-to-leaf paths; tree sums; tree structure validation	 Path Sum #112	 Path Sum II #113	 Sum Root to Leaf Numbers #129	 Path Sum III #437	 Binary Tree Maximum Path Sum #124
9	Two Heaps	Stream of numbers; median; find elements from two halves simultaneously	 Find Median from Data Stream #295	 Sliding Window Median #480	 IPO #502	 Maximum Performance of a Team #1383	 Minimize Deviation in Array #1675
10	Subsets / Backtracking	Enumerate all combinations, permutations, or power sets	 Subsets #78	 Subsets II #90	 Permutations #46	 Combination Sum #39	 Palindrome Partitioning #131
11	Binary Search	Sorted input; "find X" in O(log n); search on answer space	 Binary Search #704	 Find Minimum in Rotated Sorted Array #153	 Search in Rotated Sorted Array #33	 Find Peak Element #162	 Median of Two Sorted Arrays #4
12	Top K Elements (Heap)	Kth largest/smallest; top K frequent; K closest	 Kth Largest Element in an Array #215	 Top K Frequent Elements #347	 K Closest Points to Origin #973	 Top K Frequent Words #692	 Find K Pairs with Smallest Sums #373
13	K-way Merge	Merge K sorted arrays/lists; kth smallest across K sorted structures	 Merge k Sorted Lists #23	 Kth Smallest Element in a Sorted Matrix #378	 Smallest Range Covering Elements from K Lists #632	 Find K Pairs with Smallest Sums #373	 K-th Smallest Prime Fraction #786

14	Dynamic Programming — 1D	Optimal substructure in sequences; counting paths; max/min over choices	🟢 Climbing Stairs #70	🟡 House Robber #198	🟡 Longest Increasing Subsequence #300	🟡 Coin Change #322	🟡 Word Break #139
15	Dynamic Programming — 2D / Interval	Grid paths; string matching; subset sum; knapsack	🟡 Unique Paths #62	🟡 Longest Common Subsequence #1143	🟡 Partition Equal Subset Sum #416	🔴 Edit Distance #72	🔴 Burst Balloons #312
16	Topological Sort (Graph)	Dependency ordering; detect cycle in directed graph; course scheduling	🟡 Course Schedule #207	🟡 Course Schedule II #210	🟡 Find Order of Characters (Alien Dictionary) #269	🟡 Minimum Height Trees #310	🔴 Sequence Reconstruction #444
17	Graph BFS / DFS	Connected components; shortest path (unweighted); island problems	🟡 Number of Islands #200	🟡 Clone Graph #133	🟡 Max Area of Island #695	🟡 Pacific Atlantic Water Flow #417	🔴 Word Ladder #127
18	Union Find (Disjoint Set)	Dynamic connectivity; grouping elements; cycle detection in undirected graph	🟡 Number of Provinces #547	🟡 Redundant Connection #684	🟡 Accounts Merge #721	🟡 Number of Operations to Make Network Connected #1319	🔴 Smallest String With Swaps #1202
19	Monotonic Stack	Next greater/smaller element; histogram areas; span problems	🟢 Next Greater Element I #496	🟡 Daily Temperatures #739	🟡 Online Stock Span #901	🔴 Largest Rectangle in Histogram #84	🔴 Trapping Rain Water #42
20	Prefix Sum / Hash Map	Subarray sum equals K; count subarrays with property; running totals	🟡 Subarray Sum Equals K #560	🟡 Continuous Subarray Sum #523	🟡 Subarray Sums Divisible by K #974	🟡 Path Sum III #437	🟡 Count Number of Nice Subarrays #1248
21	Trie (Prefix Tree)	String search with prefix matching; autocomplete; word dictionaries	🟡 Implement Trie (Prefix Tree) #208	🟡 Design Add and Search Words Data Structure #211	🟡 Replace Words #648	🟡 Index Pairs of a String #1065	🔴 Word Search II #212

Study Order Recommendation

Week 1-2 → Two Pointers · Sliding Window · Fast & Slow Pointers · Binary Search
Week 3-4 → Tree BFS · Tree DFS · Merge Intervals · Cyclic Sort
Week 5-6 → Top K Elements · Two Heaps · K-way Merge · Monotonic Stack
Week 7-8 → Subsets/Backtracking · DP 1D · DP 2D
Week 9-10 → Graph BFS/DFS · Topological Sort · Union Find · Trie · Prefix Sum

Pattern Recognition Cheatsheet

Signal in the problem	Most likely pattern
Sorted array + find pair/triplet	Two Pointers
Contiguous subarray + sum/length constraint	Sliding Window
Linked list + cycle or middle	Fast & Slow Pointers
Overlapping date/time ranges	Merge Intervals
Array with numbers 1-N, find missing/duplicate	Cyclic Sort
Reverse part of a linked list	In-place Reversal
Process tree level by level	Tree BFS
Root-to-leaf path sum	Tree DFS
Running median / two halves	Two Heaps
All combinations / permutations	Backtracking
Sorted input, $O(\log n)$ target	Binary Search
Kth largest/smallest	Top K Elements (Heap)
Merge K sorted structures	K-way Merge
Count/max/min over sequences	Dynamic Programming
Dependencies / ordering tasks	Topological Sort
Island / connected component problems	Graph BFS/DFS
Dynamic grouping / cycle in undirected graph	Union Find
Next greater / smaller element	Monotonic Stack
Subarray sum equals K	Prefix Sum + Hash Map
Prefix word search / autocomplete	Trie